

ThermoRL: Safety-Constrained Reinforcement Learning for Workload Scheduling Across Heterogeneous HPC Cooling Zones Under Thermal Stress

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Abstract—AI and HPC demand is outpacing the rate at which cooling capacity can be added, and capital expansion is neither fast nor universally affordable. Facilities increasingly run air, direct-to-chip (D2C), and immersion zones concurrently; under demand spikes, maintenance, or degraded cooling infrastructure, thermal headroom becomes the binding constraint over compute. Existing schedulers ignore heterogeneous thermal physics and degrade sharply in this regime, causing reactive and costly violations. ThermoRL introduces the Thermal Headroom Index (THI), a safety scalar comparable across cooling technologies, enforced by a Lagrangian PPO agent with independent per-zone dual variables.

In a deliberately cooling-constrained 20 MW facility at $\approx 3\times$ offered load where even an omniscient Oracle records 75.7% TVR, ThermoRL cuts Thermal Violation Rate by 14.9 pp and violations per completed job by 12% against round-robin at matched throughput, with comparable per-job energy. Ablation confirms the per-zone constraint (-10.6 pp) and carbon reward (-11.5 pp) are necessary.

Index Terms—constrained Markov decision process, cooling-constrained operation, data centre cooling, direct-to-chip and immersion cooling, heterogeneous cooling zones, HPC workload scheduling, Lagrangian optimisation, safety-constrained reinforcement learning, thermal-aware scheduling, thermal headroom

I. INTRODUCTION

The density of AI and HPC workloads has broken the homogeneous cooling assumption that most schedulers were built on. The NVIDIA GB200 NVL72 draws 120 kW per rack, a power level that air cooling cannot support. Production facilities therefore run multiple cooling technologies concurrently: air-cooled rows for CPU-bound jobs, D2C-cooled aisles for dense GPU nodes, and immersion tanks for the highest-power accelerator racks. Each technology has different thermal time constants (air $\tau = 60$ s, D2C $\tau = 15$ s, immersion $\tau = 300$ s), different capacity limits, and different degradation mechanisms like filter clogging for air, fluid contamination for immersion.

Standard job schedulers such as Slurm and PBS treat these zones as interchangeable compute pools. They ask whether nodes are free; they do not ask whether the cooling subsystem has headroom to safely accept the next job. The predictable consequence is reactive thermal management: a zone is loaded until an alarm fires, the cooling plant is driven to full output

at disproportionate energy cost, and jobs are throttled or migrated under duress. On HPE Cray EX production hardware, NodeBMC sensors have been observed at 54 °C against a 50 °C limit and RouterBMC sensors at 47 °C against a 45 °C limit, both active violations; across an 885-sample telemetry extract, a temperature-only THI proxy places 9.49% of readings inside the safety band (Sec. III). These are not edge cases; they are the normal operating mode when thermal state is invisible to the scheduler.

The core difficulty is representational: temperature, heat load, and cooling hardware degradation are measured in physically incomparable units against technology-specific limits. A D2C zone at 40 °C is near its 45 °C limit; the same reading in an immersion tank with a 50 °C limit is a healthy operating point. No existing metric allows a single decision system to compare the thermal safety margin of air-cooled, D2C, and immersion zones on a common numerical scale.

ThermoRL resolves this representational gap with the Thermal Headroom Index (THI) and builds a constrained reinforcement learning (RL) scheduler around it. THI maps three zone signals to a single number that is directly comparable across cooling technologies. A Lagrangian PPO agent then treats $\text{THI}_z \geq 0.10$ as a hard per-zone constraint enforced through per-zone dual variables, while optimising a reward that trades carbon, cooling energy, and throughput. When a zone is stressed, its Lagrange multiplier rises (monotonically under the hinge cost), and the effective penalty $\lambda_z C_z$ falls to zero only when headroom returns. No manual rule-writing is needed.

The contributions of this paper are:

- **THI**: a physics-grounded normalised scalar that makes air, D2C, and immersion zones comparable, consistent with first-order RC thermal dynamics, and directly usable as an RL constraint parameter.
- **Lagrangian PPO with per-zone dual ascent**: a constrained RL algorithm that independently enforces safety margins for each zone through separate Lagrange multipliers, enabling per-zone attribution of constraint pressure without hard-coded rules.

TABLE I
ZONE PARAMETERS AND THI LIMIT VALUES (ASHRAE TC 9.9 [15])

Zone	Racks	$Q_{\max}$	$T_{\max}$	τ	PUE	d signal
Air (Z1)	200	4,000 kW	27°C	60 s	1.45	Filter ΔP
D2C (Z2)	100	8,000 kW	45°C	15 s	1.20	Contamination
Immersion (Z3)	40	8,000 kW	50°C	300 s	1.10	Fluid index

- **Rust simulation engine:** a compiled PyO3-bound environment sustaining 209,333 steps/s ($243\times$ vs. our unvectorised Python reference), making multi-seed evaluation and lookahead inside a 1 s scheduling window practical.
- **Empirical safety gain under load:** Against round-robin (throughput-matched primary result), ThermoRL cuts TVR by 14.9 pp and violations per job by 12%; versus threshold-reactive, TVR falls 15.1 pp at fixed arrival rate, at baseline energy intensity per job ($\approx 7.03\text{--}7.09$ kg CO₂/job).
- **Production hardware validation:** On 885 Cray EX samples, a temperature-only THI proxy shows 9.49% inside $\varepsilon = 0.10$ (min THI_{temp} = -0.080), a lower bound on full-THI violations.

II. BACKGROUND AND RELATED WORK

A. Thermal-Aware Placement and RL

Moore et al. [1], Vasic et al. [2], and Rostami et al. [3] target homogeneous air-cooled facilities with raw temperature signals and cannot compare risk across cooling technologies. RL has been applied to cooling actuators [4], [5], [6] and to HPC twins or GPU throttling [7], [8]; none define a cross-zone normalised safety metric with per-zone hard constraints at placement time.

B. Constrained RL and Gap

Altman [9] and Tessler et al. [10] establish Lagrangian CMDPs; we use PPO [11] with an independent dual per cooling zone. No prior system simultaneously (i) normalises multi-modal signals across heterogeneous zones, (ii) enforces that metric as per-zone CMDP constraints, and (iii) validates it on production HPC telemetry.

III. THERMAL HEADROOM INDEX

A. Definition

$$\text{THI}_z(t) = 1 - \max\left(\frac{T_z(t)}{T_{\max}}, \frac{Q_z(t)}{Q_{\max}}, \frac{d_z(t)}{d_{\max}}\right), \quad (1)$$

$$\text{THI}_z \in [-0.5, 1.0].$$

T_z is zone temperature (iLO Redfish, °C), Q_z is aggregate heat load (iLO power telemetry, kW), and d_z is a physical degradation signal normalised to $[0, 1]$ (filter differential pressure for air zones, fluid contamination index for liquid zones). Each term is divided by its zone-specific rated operational limit. Table I gives the parameters used in this work.

THI semantics: THI_z = 1.0 is full headroom; = 0.0 is at rated limit; < 0 is an active violation; < $\varepsilon = 0.10$ is within 10% of the rated limit and triggers the safety constraint.

TABLE II
TEMPERATURE-ONLY THI PROXY ($1 - T/T_{\max}$) ON HPE CRAY EX PRODUCTION TELEMETRY (885 SAMPLES, 17 ENDPOINTS, 3 SENSOR CLASSES). HEAT-LOAD AND DEGRADATION TERMS ARE UNAVAILABLE IN THIS EXPORT.

Statistic	Value
Samples analysed	885
Distinct sensor endpoints	17
Mean temperature	33.9°C
Maximum temperature	54.0°C
Mean THI _{temp}	+0.317
Minimum THI _{temp}	-0.080
Samples with THI _{temp} < $\varepsilon = 0.10$	9.49%
Endpoints with ≥ 1 violation	15 of 17
Worst endpoint (RouterBMC) violation rate	50.0%

B. Why the Max Reduction Wins Over Alternatives

The max operator rather than a weighted sum or geometric mean is critical for two properties.

Safety conservatism. A zone with $T/T_{\max} = 0.50$, $Q/Q_{\max} = 0.95$, and $d/d_{\max} = 0.10$ is near heat-load saturation. A weighted average yields ≈ 0.52 , suggesting safe headroom. The max correctly yields THI = 0.05, triggering the constraint and proactive action. This is not a corner case; it is a common operating state in liquid-cooled zones where temperature lags heat load by hundreds of seconds.

Technology-agnostic self-calibration. In air-cooled zones, temperature typically reaches its limit before heat load because thermal resistance is high. In immersion zones, heat load saturates first because the dielectric fluid’s high heat capacity keeps temperature low. The max formulation is self-calibrating: each zone type’s physically binding signal dominates without any technology-specific weighting logic.

The max of differentiable functions is subgradient-differentiable, so THI can serve directly as a neural network input feature and as a Lagrangian constraint parameter without special treatment.

C. Consistency with First-Order RC Thermal Dynamics

Zone temperatures evolve as:

$$T_z^{t+1} = T_z^t + (1 - e^{-\Delta t/\tau_z})(T_z^{\text{ss}} - T_z^t) \quad (2)$$

$$T_z^{\text{ss}} = T_z^{\text{amb}} + \frac{Q_z}{Q_{\max}}(T_z^{\text{max}} - T_z^{\text{amb}})$$

As $Q_z \rightarrow Q_z^{\text{max}}$, steady-state temperature approaches T_z^{max} and both the Q and T terms of THI approach 1 simultaneously. THI therefore provides a *leading indicator* of thermal limit approach based on current load, before temperature has risen to steady state. For the immersion zone with $\tau = 300$ s and $\Delta t = 300$ s, one step is one full time constant ($1 - e^{-1} \approx 63\%$ of steady state). At the same Δt , air ($\tau = 60$ s) reaches $1 - e^{-5} \approx 99.3\%$ and D2C ($\tau = 15$ s) $\approx 100\%$ of steady state in a single step—both are effectively memoryless—so THI’s leading-indicator property is most pronounced in immersion.

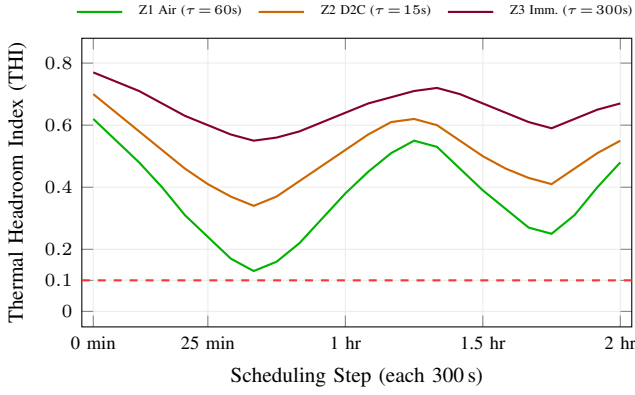


Fig. 1. Illustrative per-zone THI behaviour over a 2-hour low-load window (24 steps at $\Delta t = 300$ s), showing the qualitative dynamics the constraint is designed to maintain: under reduced load all three zones remain at or above $\varepsilon = 0.10$ (red dashed). Curves are schematic and are not an exported simulation trace. Table IV reports full 24 h episodes ($T = 288$) under sustained oversubscription, where all policies—including the Oracle—violate frequently. Figure and table are different regimes. The volatility ordering drawn reflects per-rack capacity (air 20 kW, D2C 80 kW, immersion 200 kW), not τ : at $\Delta t = 300$ s both air and D2C are effectively memoryless (Sec. III).

D. Validation on Production Telemetry

Table II summarises a temperature-only THI proxy computed from real HPE Cray EX telemetry recorded in xname/Type/Timestamp/Topic/Value format. Across 885 samples from 17 sensor endpoints, 9.49% fall inside the $\varepsilon = 0.10$ safety band and 15 of 17 endpoints record at least one violation, with the worst endpoint inside the band half the time. None of these conditions would be detectable by a scheduler watching only utilisation or job queue depth. This confirms that $\varepsilon = 0.10$ is not conservative: production systems regularly operate within 10% of rated limits. This export contains temperature only, so we evaluate the temperature term of Eq. (1) in isolation. Because THI takes the maximum over three normalised ratios, the temperature-only value is an *upper bound* on THI, and the 9.49% figure is correspondingly a *lower bound* on the true violation rate: the full metric can flag additional conditions but never fewer. Validating the heat-load and degradation terms requires CDU/BMS instrumentation and is future work.

IV. THERMORL ALGORITHM

A. CMDP Formulation

Fig. 2 shows the ThermoRL scheduling pipeline. At each 300 s step, zone telemetry and grid carbon intensity are collected, the THI vector is computed, the 9-dimensional state is formed, the Lagrangian PPO policy selects an action, and the placement is executed. After each episode, per-zone constraint violations update the three Lagrange multipliers independently.

We formulate workload scheduling as a Constrained Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \{C_z\}_{z=1}^3, \varepsilon, \gamma)$ [9].

State. Each scheduling step observes:

$$s_t = [\text{THI}_1, \text{THI}_2, \text{THI}_3, \hat{c}_t, u_1, u_2, u_3, \hat{q}_p, \hat{q}_d] \in \mathbb{R}^9 \quad (3)$$

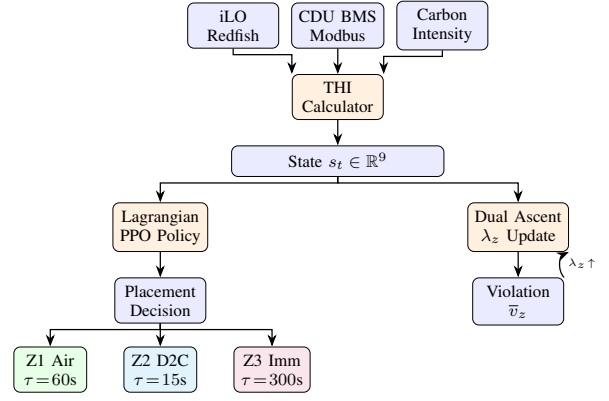


Fig. 2. ThermoRL pipeline. Telemetry feeds THI to form the 9-D state. Lagrangian PPO selects placements; per-zone violations drive independent λ_z updates (curved arc).

where $\hat{c}_t \in [0, 1]$ is normalised grid carbon intensity, $u_z \in [0, 1]$ is zone utilisation, $\hat{q}_p$ is normalised queue power demand, and $\hat{q}_d$ is normalised queue depth.

Actions. The agent selects one of five discrete actions per pending job: assign to Z1 (air), assign to Z2 (D2C), assign to Z3 (immersion), defer to next step, or throttle the highest-load job in the most-stressed zone.

Reward. The unconstrained multi-objective reward is:

$$R_t = -\alpha \cdot \hat{c}_{\text{carbon}} - \beta \cdot \hat{E}_{\text{cool}} + \gamma_R \cdot \hat{J}_{\text{completed}} \quad (4)$$

where α, β, γ_R balance carbon, cooling energy, and throughput (Eq. (4) is the conceptual multi-objective form; Section V states the numerical weights and the per-assignment implementation used in code). Carbon cost accounts for grid carbon intensity at job execution; cooling energy reflects zone PUE overhead.

Safety constraints. Each zone incurs a hinge violation cost:

$$C_z(s_t) = \max(0, \varepsilon - \text{THI}_z(t)), \quad C_z(s_t) = 0 \Leftrightarrow \text{THI}_z(t) \geq \varepsilon = 0.10 \quad \forall z, t. \quad (5)$$

The constraint is active whenever the zone is within 10% of any rated limit ($C_z > 0$).

Objective.

$$\max_{\pi} \mathbb{E}_{\pi}[\sum_t \gamma^t R_t] \quad \text{s.t.} \quad \mathbb{E}_{\pi}[\bar{C}_z] \leq 0 \quad \forall z, \quad (6)$$

where $\bar{C}_z$ is the mean per-zone violation over a rollout.

B. Lagrangian Relaxation and Dual Ascent

We relax the per-zone constraints into the objective using independent Lagrange multipliers $\lambda_z \geq 0$ [10]:

$$R_t^{\text{aug}} = R_t - \sum_{z=1}^3 \lambda_z \cdot C_z(s_t) \quad (7)$$

The policy maximises $\mathbb{E}[\sum_t R_t^{\text{aug}}]$. After each rollout of T steps, each multiplier is updated by:

$$\lambda_z \leftarrow \max(0, \lambda_z + \eta_{\lambda} \cdot \bar{v}_z), \quad \bar{v}_z = \frac{1}{T} \sum_t C_z(s_t). \quad (8)$$

with dual step size $\eta_\lambda = 0.01$ and multipliers initialised to $\lambda_z = 1.0$.

The dual update is an accumulating, zone-specific safety price. Because C_z is a hinge cost ($C_z \geq 0$), λ_z is monotonically non-decreasing within a training run, rising fastest for the zone that violates most frequently. The *effective* per-step penalty $\lambda_z C_z(s_t)$ is nevertheless time-varying within an episode: $C_z(s_t)$ falls to zero whenever a zone regains headroom, so a recovered zone incurs no penalty even though its accumulated price remains high. This prices thermal risk per-zone without any manually tuned threshold rules. Two-sided multipliers that decay when a zone is persistently safe would require a signed constraint signal and are left to future work. Under standard assumptions on the CMDP Lagrangian (compact action space, Lipschitz reward, bounded constraints) *and a non-empty feasible set*, the primal-dual iteration converges to a stationary point of the Lagrangian [10]; Sec. VII notes that the load regime evaluated here does not satisfy the feasibility condition.

C. Actor-Critic Architecture

$$\begin{aligned}\pi_\theta(a | s) &= \text{Softmax}(\text{MLP}_\theta^\pi(s)), \\ V_\phi(s) &= \text{MLP}_\phi^V(s).\end{aligned}\quad (9)$$

Both networks are two-layer MLPs with 64 hidden units and ReLU activations. The actor maps 9-dimensional state to logits over 5 actions; the critic maps it to a scalar state value. Networks share no weights. Both are trained with Adam at learning rate 3×10^{-4} .

D. PPO Update with Generalised Advantage Estimation

Within each rollout, advantages are estimated via Generalised Advantage Estimation (GAE):

$$\hat{A}_t = \sum_{k=0}^{T-t-1} (\gamma \lambda_{\text{GAE}})^k [R_{t+k}^{\text{aug}} + \gamma V(s_{t+k+1}) - V(s_{t+k})] \quad (10)$$

with $\gamma = 0.99$ and $\lambda_{\text{GAE}} = 0.95$, and with R_t^{aug} from Eq. (7). The PPO clip objective [11] is:

$$\mathcal{L}^{\text{clip}} = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (11)$$

with $\epsilon = 0.2$ and 4 update epochs per rollout.

E. Algorithm Intuition and TAWS

If Z1 is stressed, $\bar{v}_1 > 0$ raises λ_1 ; routing elsewhere drives $C_1 \rightarrow 0$ so the penalty vanishes while λ_1 stays elevated. The effective penalty $\lambda_z C_z(s_t)$ is continuous and time-varying within an episode even though λ_z updates only between episodes. At $\Delta t = 300$ s this is most useful in immersion (Sec. III). Versus TAWS [8] (per-GPU throttle under one cooling type), ThermoRL places jobs across heterogeneous zones with independent duals. Algorithm 1 summarises training.

Algorithm 1 ThermoRL: Lagrangian PPO Training

```

1: Init  $\pi_\theta, V_\phi$ ; set  $\lambda_z \leftarrow 1.0$  for  $z \in \{1, 2, 3\}$ 
2: for episode = 1 to  $N$  do
3:   Reset env; observe  $s_0$ ; clear buffer  $\mathcal{B}$ 
4:   for step  $t = 0$  to  $T - 1$  do
5:      $(a_t, \log \pi_t, v_t) \leftarrow \pi_\theta(s_t)$ 
6:      $(s_{t+1}, r_t) \leftarrow \text{env.step}(a_t)$ 
7:      $C_z^t \leftarrow \max(0, \varepsilon - \text{THI}_z(s_t)) \ \forall z$            {per-zone violations}
8:      $R_t^{\text{aug}} \leftarrow r_t - \sum_z \lambda_z C_z^t$            (Eq. (7))
9:     Store  $(s_t, a_t, R_t^{\text{aug}}, \log \pi_t, v_t, \{C_z^t\})$  in  $\mathcal{B}$ 
10:     $s_t \leftarrow s_{t+1}$ 
11:   end for
12:   Compute GAE  $\hat{A}_t$  from  $\{R_t^{\text{aug}}\}$  via Eq. (10)
13:   for epoch = 1 to 4 do
14:     Update  $\theta, \phi$  via PPO clip (Eq. (11))
15:   end for
16:   for  $z = 1$  to 3 do
17:      $\bar{v}_z \leftarrow \frac{1}{T} \sum_t C_z^t$ 
18:      $\lambda_z \leftarrow \max(0, \lambda_z + \eta_\lambda \bar{v}_z)$            (Eq. (8))
19:   end for
20: end for

```

TABLE III
SIMULATION ENGINE BENCHMARK (5,000 STEPS, MEDIAN OVER 10 RUNS)

Metric	Python	Rust	Speedup
Step latency median (μs)	1,069	4.4	243×
Step latency p95 (μs)	2,553	7.6	336×
Step latency p99 (μs)	4,308	8.7	495×
Throughput (steps/s)	781	209,333	268×
Peak memory (MB)	0.1	0.05	1.98×

F. Rust Simulation Engine

The data-centre environment is implemented in Rust using PyO3 0.21 bindings compiled with Maturin 1.5, exposing a drop-in Gymnasium interface. The engine models RC thermal dynamics (Eq. (2)), Poisson-arrival workload queues with configurable arrival rate, the THI computation for all three zones, and carbon signal interpolation from a preloaded diurnal trace.

The compiled engine reaches 209,333 steps/s versus 781 steps/s for our reference Python implementation (Table III; 243× on median latency). The Python baseline is straightforward object-oriented code and is *not* vectorised; the ratio is the practical speedup for this codebase, not an intrinsic language comparison against optimised NumPy. The operationally significant fact is absolute throughput: at 209k steps/s, multi-seed evaluation and Monte Carlo lookahead inside a 1 s window ($\sim 2 \times 10^5$ rollouts/s) become feasible.

V. EXPERIMENTAL EVALUATION

A. Environment and Training Setup

The simulation models a 20 MW heterogeneous data centre (Table I; Q_{max} sums to 4,000 + 8,000 + 8,000 = 20,000 kW).

Workloads arrive as a Poisson process at rate 1.0 jobs/s (300 jobs per 300 s step off-peak), scaled by $2.5\times$ during 08:00–18:00. Each step covers 300 s over a 24-hour episode ($T = 288$ steps; one diurnal carbon cycle). With mean job power 31.1 kW and mean duration 1998 s, Little’s-law offered load is ≈ 62.1 MW off-peak (≈ 155 MW at peak) against 20 MW capacity, a saturated regime where even an omniscient scheduler cannot keep all zones safe. Grid carbon intensity follows a 24-hour diurnal trace normalised to $[0, 1]$ with peak carbon at 08:00–10:00 and clean-energy troughs at 00:00–04:00. Workload parameters are drawn from distributions calibrated against LAST Lassen [12] and the Alibaba GPU cluster trace [13].

Thermal Violation Rate (TVR) is the fraction of (zone, step) pairs with $\text{THI}_z(t) < \varepsilon = 0.10$:

$$\text{TVR} = \frac{1}{T \cdot Z} \sum_{t=1}^T \sum_{z=1}^Z \mathbf{1}[\text{THI}_z(t) < \varepsilon], \quad (12)$$

with $Z = 3$ and $T = 288$. Ambient temperatures: air 18 °C, D2C 28 °C, immersion 30 °C. $Q_{\max,z} = \text{racks} \times \text{kW/rack}$. Simulated degradation d_z is a stress-accumulation proxy (not live sensor feeds): with $s = (T - T^{\text{amb}})/(T_{\max} - T^{\text{amb}})$, if $s > 0.85$ then d rises proportional to $(s - 0.85)\Delta t$, else it slowly recovers ($d \in [0, 1]$). Evaluation uses $\alpha = 1.0$, $\beta = 0.5$, $\gamma_R = 2.0$. The implemented per-step reward is

$$r_t = \sum_{j \in \mathcal{A}_t} (\gamma_R - \alpha \cdot c_j / c_{\text{ref}}) - \beta \cdot 0.001 \cdot E_t^{\text{cool}} + 0.5 \cdot \frac{1}{Z} \sum_z \max(0, \text{THI}_z(t) - \varepsilon), \quad (13)$$

where $\mathcal{A}_t$ is the set of jobs assigned at step t , c_j is the carbon cost of job j , $c_{\text{ref}} = 8$ kg normalises per-job carbon so the assignment term stays net-positive under typical intensities, and E_t^{cool} is step cooling energy (kWh). Eq. (4) is the conceptual objective; Eq. (13) is the form evaluated in code. The mean-THI shaping term is retained from an earlier design and is present in all variants, including the no-Lagrangian ablation. Threshold $\varepsilon = 0.10$ matches the Cray EX temperature-only proxy, where 9.49% of samples fall inside the band.

The ThermoRL agent was trained for 80 episodes using the Rust backend. All evaluation runs use the Python environment for interpretability and are averaged over 3 independent episodes with different random seeds.

B. Baselines

We compare against four schedulers spanning the range from naive to optimal:

Round-Robin (RR): Jobs rotate across zones ignoring thermal state and carbon intensity. RR is our proxy for thermally unaware production allocation; a full FCFS/backfill comparison against Slurm-class queue policies is left to future work.

Carbon-Only (CO): Jobs route to the zone with the lowest carbon-weighted energy cost, computed as $\hat{c}_t \times P_z^{\text{job}} \times \text{PUE}_z$. Thermal constraints are ignored. This represents a pure energy/carbon optimiser in the style of GECCO-class schedulers.

Threshold-Reactive (TR): Jobs assign to the zone with highest available capacity. A zone is excluded from candidacy once its temperature exceeds 90% of $T_{\max}$, and re-enters when it drops below 80%. This is representative of alarm-based production schedulers.

Oracle LP (OLP): A linear programme with perfect knowledge of future arrivals and carbon trace for the full 24-hour episode. It minimises carbon subject to thermal constraints. This bound is not deployable but establishes the achievable optimum under perfect lookahead.

C. Head-to-Head Results

Table IV reports absolute totals and per-completed-job normals. Absolute carbon/cooling favour ThermoRL largely because policies complete different amounts of work; per-job carbon is 7.03–7.09 kg across *every* scheduler, including the Oracle.

vs. Round-Robin (RR). ThermoRL reduces TVR by 14.9 pp (99.4 $\rightarrow$ 84.5) and violations per job by 12% (0.0500 $\rightarrow$ 0.0438). Carbon and cooling per job are essentially identical (7.06 kg; 4.70 vs. 4.72 kWh). RR has no thermal awareness.

vs. Carbon-Only (CO). ThermoRL reduces TVR by 8.8 pp (93.3 $\rightarrow$ 84.5) and violations per job by 6% (0.0467 $\rightarrow$ 0.0438). On energy per job, CO is slightly better (carbon 7.03 vs. 7.06 kg; cooling 4.56 vs. 4.70 kWh); the unconstrained carbon-greedy policy does *not* lose on carbon once throughput is equalised. CO’s failure mode is safety under clean-energy windows (Table VI: 290 immersion violations vs. ThermoRL’s 140), not worse carbon intensity per unit work.

vs. Threshold-Reactive (TR). ThermoRL reduces TVR by 15.1 pp (99.6 $\rightarrow$ 84.5). Absolute carbon/cooling cuts of 13.2%/13.3% are almost entirely throughput artifacts (TR completes 19,284 jobs vs. 16,664). Per job, carbon matches (7.03 vs. 7.06 kg) and violations differ by only $\approx 2\%$ (0.0447 vs. 0.0438)—inside noise at $N = 3$. The defensible claim is lower TVR under the same arrival process, not lower energy per job.

vs. Oracle LP (OLP). ThermoRL trails OLP by 8.8 pp on TVR (84.5 vs. 75.7); per job the gap collapses (carbon 7.06 vs. 7.09 kg; cooling both 4.70 kWh; viol./job 0.0438 vs. 0.0418). OLP has perfect knowledge of the next 24 h. Residual Oracle TVR under omniscience is evidence the load regime is infeasible for any policy (Section VII).

D. What Survives Fair Normalisation

TVR and violations-per-job answer different operational questions. Under a fixed arrival process, an operator experiences the wall-clock fraction of time zones spend inside the safety band—that is TVR, and it drives cooling-plant stress and hardware exposure regardless of how much work completed. Violations-per-job measures safety efficiency per unit of delivered service. We report both. Against threshold-reactive, ThermoRL improves TVR by 15.1 pp but only $\approx 2\%$ on violations-per-job, because TR completes 16% more work at proportionally similar risk; we therefore claim only the former against TR. Against round-robin, throughput is nearly

TABLE IV

SCHEDULER COMPARISON (20 MW, 24 H EPISODES, 3-SEED MEAN). TVR VIA EQ. (12) ($T = 288$, $Z = 3$, $\epsilon = 0.10$). ABSOLUTE CARBON/COOLING ARE *not* THROUGHPUT-MATCHED; PER-JOB COLUMNS ARE THE FAIR COMPARISON. ARRIVAL RATE 1.0 JOBS/S YIELDS SUSTAINED OVERSUBSCRIPTION (ORACLE TVR 75.7%); READ AS RELATIVE RANKINGS UNDER SATURATION. ABSOLUTE CARBON AND COOLING ARE SHOWN UNBOLDED BECAUSE THEY ARE NOT THROUGHPUT-MATCHED; **BOLD** MARKS THE BEST DEPLOYABLE VALUE IN THROUGHPUT-NORMALISED AND RATE COLUMNS ONLY (ORACLE EXCLUDED; LOWER BETTER EXCEPT JOBS).

Scheduler	TVR (%)	Carbon (kg)	Cooling (kWh)	Jobs	kg/job	kWh/job	Viol./job
ThermoRL (ours)	84.5	117,666	78,316	16,664	7.06	4.70	0.0438
Round-Robin	99.4	121,211	80,979	17,172	7.06	4.72	0.0500
Carbon-Only	93.3	121,249	78,692	17,252	7.03	4.56	0.0467
Threshold-Reactive	99.6	135,474	90,321	19,284	7.03	4.68	0.0447
Oracle LP (bound)	75.7	110,872	73,438	15,631	7.09	4.70	0.0418

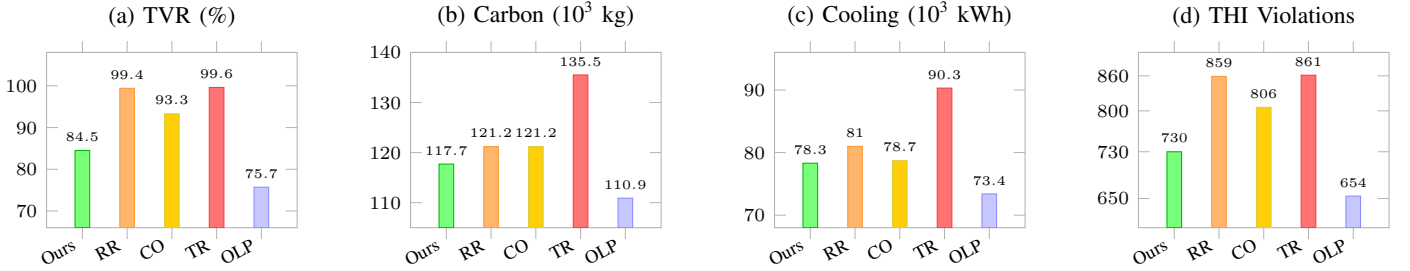


Fig. 3. Scheduler comparison across all four metrics (lower is better for TVR, carbon, and cooling). Ours = ThermoRL ■, RR = Round-Robin ■, CO = Carbon-Only ■, TR = Threshold-Reactive ■, OLP = Oracle LP ■ (non-deployable bound). Absolute totals; see Table IV for per-job normals.

matched (17,172 vs. 16,664 jobs, a 3% difference) and both metrics agree: 14.9 pp lower TVR and 12% fewer violations per job. The round-robin comparison is therefore the throughput-matched result, and is the one we regard as primary.

The three deployable baselines fail on *safety*, not on energy intensity per job: RR is thermally blind (highest viol./job, 0.0500); CO over-routes to immersion in clean-energy windows (Table VI) while achieving the best carbon and cooling per job; TR is reactive and posts near-100% TVR but matches ThermoRL on kg/job because it completes more work. ThermoRL’s reduction in violations per unit work versus RR (−12%) and CO (−6%) comes at energy efficiency indistinguishable across policies (7.03–7.09 kg/job). We do *not* claim a carbon win over a dedicated carbon-greedy scheduler once throughput is equalised.

E. Ablation Study

Table V isolates each algorithmic component. The no-constraint variant freezes $\lambda_z \equiv 0$, disabling dual ascent. The no-carbon variant removes $\hat{c}_{\text{carbon}}$ from the reward. The high-carbon variant doubles α .

Freezing $\lambda_z \equiv 0$ raises TVR +10.6 pp even with mean-THI shaping still on, isolating *per-zone* pricing rather than thermal

TABLE V

ABLATION: EACH COMPONENT’S CONTRIBUTION TO TVR. ΔTVR IS VS. FULL THERMORL WITHIN THIS TABLE (POSITIVE = MORE VIOLATIONS). Ablations use 3-hour episodes ($T = 36$) to reduce compute cost; absolute values are not comparable to Table IV (24 h, $T = 288$). ONLY WITHIN-TABLE ΔTVR IS INTERPRETABLE.

Variant	TVR (%)	Carbon (kg)	Cooling (kWh)	ΔTVR (pp)
Full ThermoRL	76.9	16,313	9,272	—
No Lagrangian ($\lambda = 0$)	87.5	18,263	9,628	+10.6
No carbon reward	88.4	18,476	10,079	+11.5
High carbon ($2\times$)	83.3	18,195	9,710	+6.4
Round-Robin (reference)	91.2	17,907	9,787	+14.3
Oracle LP (upper bound)	72.2	16,788	8,726	−4.7

awareness in general. Removing the carbon term raises TVR +11.5 pp (over-routing to low-PUE immersion); doubling α raises it +6.4 pp. Full ThermoRL captures 75% of the Oracle–RR TVR gap in this table ($((91.2 - 76.9) / (91.2 - 72.2))$).

F. Per-Zone Violation Breakdown

Table VI shows the distribution of thermal violations across the three zones. Air (Z1) generates the most violations in all

TABLE VI
PER-ZONE VIOLATION COUNTS (24 H EPISODES, 3-SEED MEAN; SAME
REGIME AS TABLE IV). TOTALS CORRESPOND TO THE VIOLATION COUNTS
UNDERLYING TVR IN TABLE IV AND PLOTTED IN FIG. 3(D).

Scheduler	Z1 Air	Z2 D2C	Z3 Imm.	Total
ThermoRL (ours)	389	201	140	730
Round-Robin	471	237	151	859
Carbon-Only	318	198	290	806
Threshold-Reactive	480	241	140	861
Oracle LP (bound)	348	178	128	654

schedulers because its low per-rack capacity (20 kW) and high PUE (1.45) make it the first zone saturated under load. ThermoRL reduces Z1 violations by 17.4% versus RR (471 $\rightarrow$ 389) and cuts the Z3 (immersion) spike that Carbon-Only produces by over-routing to the low-PUE immersion zone during clean-energy windows (290 $\rightarrow$ 140).

The Carbon-Only scheduler’s Z3 violations (290 vs. ThermoRL’s 140) illustrate the failure mode precisely: CO routes aggressively to immersion during low-carbon hours because immersion has the lowest PUE overhead, saturating it thermally. ThermoRL’s accumulated Z3 multiplier, combined with the instantaneous violation cost $C_3(s_t)$, prices the safety violation back into the placement decision at those steps.

After 80 training episodes the per-zone multipliers reach $\lambda = [1.054, 1.052, 1.037]$. Because the dual update is monotone under a hinge cost these are *accumulated* prices rather than converged steady-state values, and their ordering—air highest—reflects cumulative violation exposure, consistent with Table VI. The spread is small (1.6% of the initial value after 80 episodes): the duals separate in the expected order but are far from converged, consistent with the limited training budget (Sec. VII). Demonstrating that the per-zone decomposition outperforms a single scalar multiplier requires longer training and is future work. Per-episode λ_z trajectories were not logged. The formulation is nonetheless motivated by an attribution argument: a single aggregate constraint produces a scalar penalty that cannot identify which zone is the current safety bottleneck at any given step.

VI. IMPLEMENTATION AND DEPLOYMENT

A. Software Stack and Deployment

Python: Gymnasium 0.29, PyTorch 2.x. Rust/PyO3 on macOS and Linux (Table III). Workloads follow LAST/Alibaba-calibrated distributions [12], [13]; carbon is the Sec. V diurnal trace (Electricity Maps [14] is a drop-in). Deployed state would use iLO Redfish (T_z, Q_z), CDU BMS (d_z ; stress proxy in simulation), and an external carbon feed; $\Delta t=300$ s matches HPCM. At 209k steps/s, multi-seed eval and $\sim 2 \times 10^5$ rollouts/s fit a 1 s window.

VII. DISCUSSION

When ThermoRL matters. Thermal-aware placement only differentiates when thermal constraints actively bind. Where

cooling capacity comfortably exceeds offered load, headroom is abundant, every zone accepts every job, and allocators that ignore thermal state perform much like those that do not: there is little for a constrained policy to optimise. The regime evaluated here is the opposite one. At $\approx 3.1 \times$ offered load off-peak and $\approx 7.8 \times$ at peak, cooling is the scarce resource, the feasible set of Eq. (6) is empty, and every allocator—including the Oracle—spends most zone-steps inside the safety band. Our results characterise this regime specifically: under identical overload, pricing per-zone headroom reduces violation mass relative to blind and reactive allocation. We make no claim about regimes in which constraints rarely bind; identifying the load threshold at which thermal-aware placement begins to pay is left to future work.

Practical deployment model. This suggests a two-regime operational model rather than a replacement for classical scheduling. Under normal load, conventional queue and backfill policies are adequate and impose no thermal reasoning overhead. As headroom erodes—during demand spikes, seasonal ambient peaks, or reduced cooling availability—placement escalates to a thermal-constrained policy that prices zones by remaining headroom, reverting once headroom recovers. THI is a natural trigger for such a switch precisely because it is comparable across cooling modalities: one threshold carries the same physical meaning in an air row, a D2C aisle, and an immersion tank. We present this as deployment intuition, not as a measured result. We did not implement or evaluate a load-adaptive controller; the switching rule, hysteresis, and stability of such a scheme are open systems questions.

Demand growth and capacity degradation. Cooling-constrained operation is not an exotic corner case, and it does not require workload growth alone to arise. Rack power density is rising faster than facility cooling is refreshed—a single GB200 NVL72 rack draws 120 kW—so the ratio of offered thermal load to installed cooling capacity trends upward even at constant job count. Capacity can also *fall*: CDU or pump faults, chiller maintenance, filter fouling, coolant degradation, and partial loop isolation all reduce effective cooling while work continues to arrive, and the resulting thermal stress accelerates the degradation that caused it. A facility comfortable at full capacity can therefore enter the regime studied here through a maintenance window alone, with no change in demand. Because capital expansion is slow, operators facing either trend must place work more carefully inside a fixed envelope rather than enlarge the envelope—the problem this paper addresses.

Limitations. The evaluation arrival rate of 1.0 jobs/s places the 20 MW facility in sustained oversubscription (offered load $\approx 3.1 \times$ capacity off-peak, $\approx 7.8 \times$ at peak; Oracle LP TVR 75.7%). Because $C_z \geq 0$, the constraint of Eq. (6) requires $\mathbb{E}[C_z] = 0$; under this load regime that set is empty, as the Oracle’s 75.7% TVR confirms. The primal-dual iteration therefore does not converge to a feasible point: λ_z accumulates monotonically and the policy minimises violation mass rather than eliminating it. Feasibility analysis under a calibrated load regime, where the constraint set is non-empty, is future work. Absolute TVR is not an operational

safety rate; only pairwise differences under the same load are meaningful. Absolute carbon/cooling in Table IV are not throughput-matched (15,631–19,284 jobs); per-job normals show comparable energy efficiency, with ThermoRL’s primary gain in violations per unit work against the throughput-matched round-robin baseline. Enforcing throughput as an explicit CMDP constraint is future work. Results are means over $N = 3$ evaluation seeds; we report no standard deviations or confidence intervals, so the reported TVR gaps are directionally stable but not statistically tested effect sizes. Our model uses constant per-zone PUE and does not represent the efficiency penalty a real plant incurs near its limit, so the measured cooling difference against threshold-reactive scheduling is a throughput artifact rather than evidence of that mechanism; load-dependent PUE is future work. The agent was trained for 80 episodes and no training-budget sweep was performed; the residual gap to the Oracle may therefore reflect undertraining as well as absent lookahead. The action space includes a throttle action alongside the three placement actions and deferral; its usage frequency was not logged, so the decomposition of ThermoRL’s reduced completion count into deferral versus throttling is unavailable. Restricting the action space to placement-only is future work. Workloads are synthetic Poisson arrivals, not replayed HPC traces. Validation against public traces is left to future work. The carbon trace is synthetic; live Electricity Maps integration would add forecast uncertainty that the current state vector does not represent. At $\Delta t = 300$ s, air and D2C are memoryless (Sec. III).

VIII. CONCLUSION

ThermoRL demonstrates that proactive safety-constrained reinforcement learning can reduce thermal violations under heterogeneous HPC cooling relative to reactive and thermally blind baselines. The two core algorithmic contributions are the Thermal Headroom Index, which provides a physics-grounded, technology-agnostic, and constraint-ready safety representation, and Lagrangian PPO with per-zone dual ascent, which enforces that safety representation as a hard limit while co-optimising carbon and energy.

On a 20 MW three-zone simulation under sustained over-subscription, the throughput-matched primary result is versus round-robin: ThermoRL achieves a TVR of 84.5% against 99.4% (−14.9 pp) and 12% fewer violations per completed job. Versus threshold-reactive, TVR falls 15.1 pp at fixed arrival rate (99.6 → 84.5); versus carbon-greedy scheduling, TVR is 93.3%. Per completed job, carbon intensity is comparable across all policies (≈ 7.03 – 7.09 kg). We do not claim a per-job carbon win over the carbon-greedy baseline. Ablation studies confirm that both the Lagrangian constraint (10.6 pp when removed) and the carbon reward (11.5 pp when removed) are necessary within the 3 h ablation regime; neither alone achieves the full benefit. The compiled Rust engine at 209,333 steps/s ($243\times$ vs. our unvectorised Python reference) makes online evaluation practical at sub-second latency.

A temperature-only THI proxy on production HPE Cray EX telemetry (885 samples, 17 endpoints) places 9.49% of readings

inside the $\varepsilon = 0.10$ band with minimum $\text{THI}_{\text{temp}} = -0.080$ —a lower bound on full-THI violations—confirming that the safety threshold corresponds to real conditions production systems regularly exceed.

Future work includes: (i) integrating 15-minute carbon intensity forecasts from Electricity Maps as an additional state feature to close the Oracle gap; (ii) connecting ThermoRL as an S-RAPS external scheduler for validation on Frontier telemetry replay; (iii) extending the constraint to degradation budget limits over multi-day horizons.

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